

Antonio Rafael Vazquez Pantoja

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EDUCATION

Texas A&M University

Bachelor of Science in Mechanical Engineering, Minor in Biomedical Engineering

May 2028

GPA: 3.4

EXPERIENCE

Biotex Inc.

May 2026 - Aug 2026

R&D Mechanical Engineering Intern

Houston, TX

- Designed and prototyped automated medical-device test fixtures for needle rotation, puncture actuation, and reset positioning, reducing manual technician involvement and supporting high-cycle verification testing.
- Developed modular SolidWorks fixture assemblies and prepared CNC-ready machined parts using fastened components, heat-set inserts, snap-fit features, and 3D-printed prototypes to accelerate R&D iteration and troubleshooting.
- Performed snap-fit strain calculations for 3D-printed needle-grabber features, guiding claw geometry, wall thickness, and material selection to prevent fracture during repeated insertion/removal cycles.
- Created engineering drawings, BOM updates, prototype documentation, and design-change records while supporting catheter-handling and UV-curing fixture redesigns in a GMP/ISO 13485 medical-device R&D environment.

Texas A&M – Biomedical Engineering Department

Jan 2026 - Present

Undergraduate Research Assistant, Ethicon (Johnson and Johnson Sponsored Research)

College Station, TX

- Conduct corrosion-mechanical coupled testing on bioabsorbable magnesium surgical staples to evaluate degradation-driven changes in structural integrity, mechanical performance, and failure behavior.
- Perform tensile stress-strain testing across varied bend geometries to identify elastic limits, yield behavior, bend-radius constraints, and failure points relevant to bioresorbable implant design.
- Analyze mechanical response under progressive corrosion conditions to support evaluation of material degradation, geometric failure modes, and design constraints for implantable medical devices.

Texas A&M – Biomedical Engineering Department

Sept 2025 - Jan 2026

Undergraduate Research Assistant, Cardiovascular Devices

College Station, TX

- Assembled RF balloon catheter prototypes for fetal hypoplastic left heart syndrome applications by soldering RF PCB components to balloon structures for benchtop and porcine tissue evaluation.
- Iterated through three test-bed designs to improve RF-induced tissue denaturing consistency in aqueous conditions, resolving tissue tearing and inconsistent electrode contact through fixture redesign.
- Redesigned test-bed geometry and integrated silicone fixturing to improve tissue positioning, RF PCB contact, load distribution, and energy-transfer consistency during device testing.

Texas A&M - Mechanical Engineering Department

June 2025 - May 2026

Undergraduate ME Engineering Research Assistant

College Station, TX

- Co-authored a peer-reviewed publication on the ANGEL soft robotic gripper; designed CAD models for primary structural components, motor spool geometry, and cable-routing architecture to improve wire tensioning and contact force distribution.
- Designed and iterated 3D-printed TPU pocket geometries, including cushioning, flexure, and contact-area variants, to reduce localized contact stress during delicate object handling.
- Conducted experimental trials across 56 tomatoes, achieving a 0% immediate damage rate and under 9% bruise rate five days after grasping.

Del Sol Foods Co.

Oct 2023 - Present

Manufacturing and Industrial Engineering Intern

Brenham, TX

- Supported manufacturing, maintenance, and quality teams by troubleshooting equipment issues, improving SOPs, and maintaining production documentation in a high-volume food manufacturing environment.
- Evaluated bakery production equipment, facility constraints, and vendor options with leadership, supporting equipment selection that reduced labor requirements by 30% and avoided \$2,000 in capital expenditure.

PROJECTS

Turtle Robotics – LARM: Laboratory Robotic Manipulator

Sept 2025 - May 2026

- Led mechanical design of a servo-actuated gripper for manipulating 500 mL laboratory beakers in wet-lab environments.
- Engineered a 3-gear transmission system to synchronize claw motion and amplify torque from a 20 kg·cm servo motor.
- Evaluated slip risk on wet glass surfaces and integrated silicone contact pads to increase friction while designing claw curvature to conform to cylindrical glassware geometry.
- Iterated SolidWorks assemblies, tolerance stack-ups, and gear-spacing layouts to improve smooth gear meshing, repeatable actuation, and manufacturability.

TECHNICAL SKILLS

Mechanical Design: SolidWorks, Inventor, Onshape, AutoCAD, GD&T/ASME Y14.5, assemblies, tolerance stack-up, engineering drawings, fixture design, mechanism design, DFM/DFA

Prototyping & Manufacturing: FDM/SLA 3D printing, heat-set inserts, CNC design preparation, fastener selection, soldering, PCB assembly, rapid prototyping

Testing & Analysis: tensile testing, stress-strain analysis, corrosion-mechanical testing, bend-radius characterization, failure-mode analysis, prototype validation

Software: Python, Excel

Languages: English (Fluent), Spanish (Fluent)